| Composition of MS (Murashige and Skoog) Medium | |
|---|---|
| **Macronutrients**: | |
| Ammonium nitrate ($NH_4NO_3$) | 1650.0 mg/l |
| Potassium nitrate (KNO3) | 1900.0 mg/l |
| Calcium chloride (CaCl2 2H2O) | 440.0 mg/l |
| Magnesium sulphate (MgSO4 6H2O) | 370.0 mg/l |
| Potassium dihydrogen phosphate (KH2PO4) | 170.0 mg/l |
| **Micronutrients**: | |
| Manganese sulphate (MnSO4 4H2O) | 22.3 mg/l |
| Zinc sulphate (ZnSO4 4H2O) | 8.6 mg/l |
| Boric acid (H3BO3) | 6.2 mg/l |
| Potassium iodide (KI) | 0.83 mg/l |
| **Minor nutrient:** | |
| Sodium molybdate (Na2 MO4 2H2O) | 0.250 mg/l |
| Cupric sulphate (CuSO4 5H2O) | 0.025 mg/l |
| Cobaltous chloride (CoCl2 6H2O) | 0.025 mg/l |
| **Iron stock** | |
| Na EDTA | 37.25 mg/l |
| Ferrous Sulphate (FeSO4 7H2O) | 27.85 mg/l |
| **Vitamins** | |
| Glycine | 2.0 mg/l |
| Nicotinic acid | 0.5 mg/l |
| Pyridoxin HCl | 0.5 mg/l |
| Thaiamine HCl | 0.1 mg/l |
| **Growth Hormones** | |
| IAA | 1.30 mg/l |
| Kinetin | 0.4–10.0 mg/l |
| Myo-inositol | 100.0 mg/l |
| Sucrose | 30.0 g/l |
| **Solidifying Agent** | |
| Agar | 8.0 g/l |

## 3. Culture condition

### pH

The pH of medium is normally adjusted between 5.6 to 6.0 for the best result.

### Temperature

The cultures should be incubated normally at constant temperature of 25°C ± 2°C for optimal growth.

### Humidity and Light Intensity

The cultures require 50-60% relative humidity and 16 hours of photoperiod by the illumination of cool white fluorescent tubes of approximately 1000 lux.

### Aeration

Aeration to the culture can be provided by shaking the flasks or tubes of liquid culture on automatic shaker or aeration of the medium by passing with filter-sterilized air.

## 4. Induction of Callus



**Figure 5.4:**
Induction of callus

Explant of 1-2 cm sterile segment selected from leaf, stem, tuber or root is inoculated (transferring the explants to sterile glass tube containing nutrient medium) in the MS nutrient medium supplemented with auxins and incubated at 25°C ± 2°C in an alternate light and dark period of 12 hours to induce cell division and soon the upper surface of explant develops into callus. Callus is a mass of unorganized growth of plant cells or tissues in *in vitro* culture medium.

## 5. Embryogenesis



**Figure 5.5:**
Embryogenesis

The callus cells undergoes differentiation and produces somatic embryos, known as **Embryoids**. The embryoids are sub-cultured to produce plantlets.



TN_GOVT_BOTANY_XII_PAGES_121-135 CH 05.indd 113 3/8/2020 2:50:42 PM